

Soccermatics Project 3

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Evaluating Passing-Decisions with Tracking Data

An Expected-Pass-Value framework on Real Madrid data

Thank you to Soccermatics for providing the SkillCorner data

1. Motivation

Almost every public football metric is **outcome-based**. From the standard headline metrics of goals, assists and passes completed, to the more technical probabilistic models of Expected Goals (xG), Expected Threat (xT; Singh, 2018), and Valuing Actions by Estimating Probabilities (VAEP; Decroos et al., 2019) – all of them model what happened on the pitch. While many of these models and metrics provide invaluable signal on a player's ability, they do not capture if the player actually made the right decision.

More concretely, a player who completes a forward pass into a team-mate under pressure, likely to lose the ball moments later, while ignoring a safer and more threatening option elsewhere, will still be rewarded by outcome-based metrics, including Expected Threat (xT). xT rewards the forward progression, but does not account for the more valuable passing option the player ignored.



Graphic: Bellingham completes a pass to Vinicius and records a positive xT added of 0.013, but is not penalised for ignoring the more valuable option to Mbappé.

Imagine pausing the match at the instant a player releases a pass. Which team-mates were available, how valuable was each option, and did the player choose the best one?

This project answers that question using a **counterfactual expected pass value (xPV) framework**. At the moment of the pass, we value every team-mate as a potential receiver and evaluate the chosen option against the alternatives that were available at that instant - independent of whether the pass came off.

The framework therefore evaluates pass selection, not whether or when the player should have passed. The report first explains how the model was implemented, illustrates its behaviour through match examples, and then aggregates the resulting scores into player-level decision metrics.

The formulation is heavily inspired by the Expected Possession Value (EPV) framework of Fernández, Bornn and Cervone (2021). We also draw on earlier pitch-control work of Fernández and Bornn (2018) and Spearman (2018). The graph-based representation of the game state draws on Sahasrabudhe and Bekkers (2023), and the project makes use of Bekkers' open-source *unravelsports* package (Bekkers, 2024). Singh's xT framework (2018) provides a comparison baseline.

2. Data

The dataset comprises 98 Real Madrid matches from SkillCorner, pairing broadcast tracking data at 10 fps with synchronised dynamic event data, and spans two full seasons of La Liga and UEFA Champions League football (2023/24 and 2024/25). Modelling is restricted to open-play passes, with set pieces excluded.

3. The model

Model components:

I – the set of all team-mates available to receive the pass, with $i \in I$ denoting an individual candidate receiver.

π_i – the probability the passer plays to team-mate i , given the player passes, with $\sum_i \pi_i = 1$. Conditioned on a pass being attempted to a team-mate, the probabilities across team-mates sum to 100%: a softmax over the candidates.

$P_{cmp,i}$ – the probability the pass to i is completed. In deployment, this is a learned base probability times an executability gate:

$$P_{cmp,i} = P_{model,i} \times g_{i,i} \quad (\text{see section 3.2})$$

$V_{cmp,i}$ – the value of the resulting state if the pass is completed, estimated from the discounted open-play xG subsequently generated in that possession

$V_{turnover,i}$ – the cost if the pass fails, estimated from the discounted open-play xG generated by the opponent immediately after the turnover.

Model Equation

For each candidate $i \in I$, the value of the pass (xPV: expected passing value) is:

$$xPV_i = \underbrace{P_{\text{cmp},i} \cdot V_{\text{cmp},i}}_{\text{upside reward}} + \underbrace{(1 - P_{\text{cmp},i}) \cdot V_{\text{turnover},i}}_{\text{downside risk}}$$

The selection model π then weights these option values into the expected value of the decision presented to the passer:

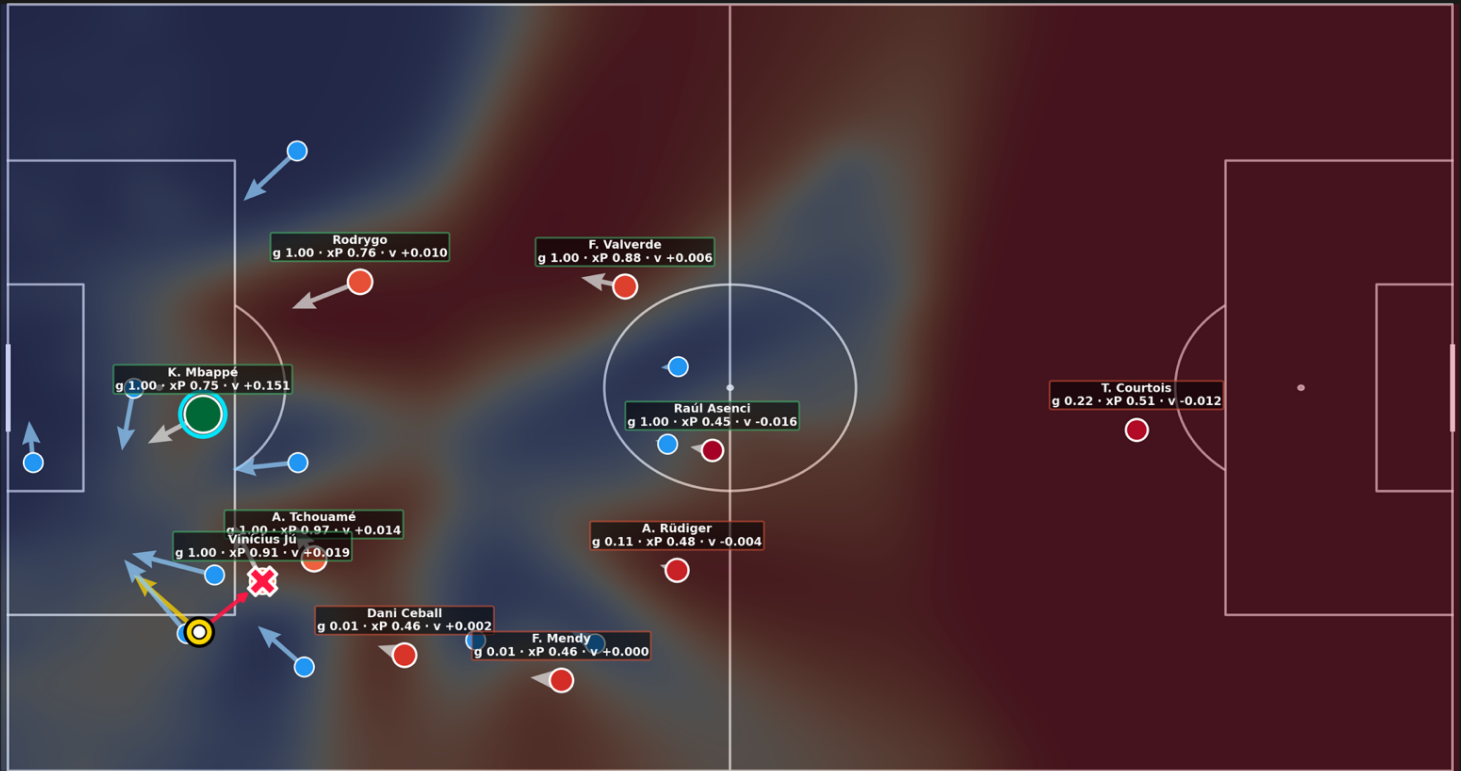
$$xPV = \sum_{i \in I} \pi_i xPV_i$$

This gives every available passing option an expected value, allowing the chosen pass to be evaluated against its alternatives.

The model in action: The cost of ignoring Mbappé

A better, catchable ball was on — J. Bellingham → Vinícius, 73:01 (RM v Man City)

Bellingham plays Vinícius (73:01) — xPV rates the ball to Mbappé (cyan ring) the better, catchable option
 bg = pitch control · gold = passer (+vel) · red = ball played · cyan ring = best legal option · cyan ★ = pass-into-space target



How each option's value is built								
$value = xPass \times V^+ + (1 - xPass) \times V^-$ $xPass = P(model) \times g$ → run rows: P, g, xPass, V ⁺ , V ⁻ all measured at the in-behind destination								
receiver	dir	π (GNN)	P model	gate g	xPass	V ⁺	V ⁻	value
K. Mbappé ○	side	63%	0.75	1.00	0.75	+0.203	-0.002	+0.151
Vinicius Jún ▶	back	18%	0.91	1.00	0.91	+0.021	-0.003	+0.019
A. Tchouamén	back	6%	0.97	1.00	0.97	+0.015	-0.003	+0.014
Rodrygo	back	12%	0.76	1.00	0.76	+0.014	-0.003	+0.010
F. Valverde	back	1%	0.88	1.00	0.88	+0.007	-0.004	+0.006
Dani Ceballo	back	0%	0.71	0.01	0.01	+0.008	-0.003	-0.003
F. Mendy	back	0%	0.55	0.01	0.01	+0.006	-0.004	-0.004
A. Rüdiger	back	0%	0.81	0.11	0.09	+0.005	-0.012	-0.010
Raúl Asencio	back	0%	0.45	1.00	0.45	+0.005	-0.033	-0.016
T. Courtois	back	0%	0.98	0.22	0.21	+0.004	-0.028	-0.021

a better, catchable option was on

vanilla xT-added (actual ball end) +0.013

situation xPV (π-weighted) = $\sum \pi \cdot v_i$ +0.10049

ball played → Vinicius Júnlor +0.01874

best legal option → K. Mbappé +0.15059

xPV value-added = played - π-weighted = -0.08175 X worse than typical

π (GNN) = deployed GNN selection model (how likely he plays each option) · ▶ chosen · ○ best legal · P model = learned completion (interception × delivery), pitch-control is one input feature
V⁺ = value if completed · V⁻ = turnover cost · →run = pass into the space the runner attacks: P, g, xPass, V⁺, V⁻ all measured at the in-behind destination; each option's value already = max(feet, run)

Bellingham's ball to Vinicius was to feet (received 2.4 m from his position) — a safe 91% ball, but a lower-value spot.
Mbappé (the model's best option) is also a catchable FEET ball — 75% completion in a far higher-value spot: the simpler, better option existed.

Note: V^+ and V^- represent $V_{cmp,i}$ and $V_{turnover,i}$ respectively.

Bellingham has the ball with several team-mates to aim at. The model rates **Mbappé the likeliest target ($\pi = 63\%$)**, followed by Vinicius (18%), Rodrygo (12%) and Tchouaméni (6%).

The potential pass to Mbappé has a 75% chance of success but is in a highly valuable goal-scoring position worth +0.203 if he can be reached, with a negligible turnover cost of −0.002, resulting in:

$$xPV_{Mbappé} = 0.75(0.203) + 0.25(-0.002) \approx 0.15$$

The pass to Vinicius is safer - a 91% chance of success – but is in a far less immediately threatening position worth only +0.02:

$$xPV_{Vini} = 0.91(0.021) + 0.09(-0.003) \approx 0.02.$$

Weighting all available options by their selection probabilities gives an overall decision-state value of approximately **0.10**.

$$(\pi_{Mbappe} \times xPV_{Mbappé} + \pi_{Vini} \times xPV_{Vini} + \dots \pi_{Courtois} \times xPV_{Courtois})$$

The chosen pass therefore records approximately **−0.08 xPV added**, despite generating positive xT.

Modelling

3.1

π_i : Probability of the passer passing to team-mate i

How this was modelled

At each pass frame, every team-mate is treated as a candidate receiver. The model scores each candidate, and a softmax converts those scores into π_i , the probability that the passer chooses that player.

Building the graph. At the pass frame, the game state is represented as a graph using `SoccerGraphConverter`, with every player and the ball as nodes. Team-mates and opponents are connected within their respective teams, with information passing between the two groups through the ball node. Because attacking and defending players are not directly connected, candidate-specific pressure information is included through node features.

Node features (15 per node). Two features locate the node on the pitch (x, y). Three describe its motion: speed, and the velocity components v_x and v_y . Two give proximity: distance to goal and distance to the ball. Four give orientation - the angle to goal and the angle to the ball, each stored as a cosine-sine pair. The last four are binary flags identifying what the node is: does the player belong to the team in possession, is the player the goalkeeper, is it the ball node, is it the ball carrier.

The ball node carries only its position and ball flag: its velocity is masked, since its vector would otherwise point straight at the receiver.

Three candidate-specific features are added:

Pitch-control completion prior: the estimated probability that the ball reaches the candidate without interception (section 3.2).

Lane congestion: the minimum defender distance from the passing lane.

Executability gate : whether the passer can realistically strike the ball toward that candidate given orientation and pressure (section 3.3).

Edge features (6 per edge). Each edge describes the relationship between the two nodes it connects, in six numbers: one for the distance between the two players, one for their speed difference, two for the bearing from one to the other (its cosine and sine), and two for the angle between their velocity vectors (cosine and sine again) - the last pair telling the network whether the two players are moving in the same or opposing directions.

Architecture:

Node features are projected into a 64-dimensional representation and passed through two CGConv message-passing layers. A small MLP then scores each candidate, with a softmax producing π_i , the probability that the passer selects candidate i . Two layers are sufficient for

information to cross between the teams through the ball node. The model is trained using masked cross-entropy on the true receiver.

A conditional-logit baseline was trained over the same candidates using eight hand-crafted spatial, pressure and passing-lane features.

Evaluation

Evaluation is held out by game, using grouped three-fold cross-validation: each match is assigned wholly to one-fold, so no frames from the same match appear on both sides of a split. This matters because consecutive frames within a match are nearly identical, which otherwise would let the model memorise situations rather than generalise, and inflate every reported score.

Model selection compared ten configurations on mean held-out AUC: each of the two feature sets (the 15 base features alone, or with the three appended features) was trained under five architecture settings, varying the number of message-passing layers, hidden width, weight decay and dropout. The selected configuration - the appended features with two message-passing layers - is the model described above and deployed throughout this report.

Results

Model	AUC $\uparrow$	Brier $\downarrow$	Top-1 $\uparrow$	Mean P(true receiver) $\uparrow$
Conditional logit (8 features)	0.88	0.65	49.9%	0.35
GNN (15 features)	0.95	0.50	63.1%	0.52
GNN (18 features)	0.96	0.47	65.5%	0.57

Adding the candidate-specific features improves performance across every reported metric. The deployed 18-feature GNN reaches 0.96 AUC, selects the true receiver on 65.5% of passes, and assigns the true receiver a mean probability of 0.57.

3.2 P_{cmp} : Pass completion probability

For a potential pass to receiver i , completion is decomposed into two components:

$$P_{cmp,i} = P_{model,i} \times g_i$$

Where $P_{model,i}$ estimates whether the ball reaches the target if the pass is struck toward it, and g_i estimates whether the passer can realistically execute that pass in the first place.

Concretely:

$P_{model,i} = P(\text{ball reaches } i \text{ un-intercepted} \mid \text{it is struck toward } i)$

$g_i \in [0,1] = P(\text{the pass can be struck toward } i \text{ at all})$

This distinction matters because attempted-pass data can train the first component, but provides almost no labelled examples of passes that were physically unavailable and

therefore never attempted. We therefore learn $P_{model,i}$ from observed passes and construct g_i from player kinematics and pressure.

Constructing $P_{model,i}$

We use a Spearman-style pitch-control model as a physics-based completion prior (Spearman et al., 2017; Spearman, 2018). Given player positions and velocities, together with an assumed ball speed and player movement model, it estimates the probability that the ball reaches the target before an opponent can intercept it.

On its own, this interception prior achieves a held-out AUC of 0.824. It does not capture every source of pass difficulty, however, so we use it as a feature in a gradient-boosted completion model alongside pass geometry, lane congestion and pressure. The resulting $P_{model,i}$ improves held-out AUC to 0.892.

Pitch control still assumes that the ball can be struck toward the target in the first place. The remaining orientation-and-pressure constraint is therefore handled by the executability gate (g), described in section 3.3.

3.3 The executability gate g

Pitch control estimates whether the ball will reach a target **if the pass is played in that direction**. It does not consider whether the passer can **realistically produce that pass from his current movement and orientation**. For example, a forward passing lane may appear open even though the passer is moving quickly towards his own goal and cannot turn before pressure arrives.

g_i estimates whether the passer can realistically strike the ball in that direction. A value close to one leaves the option unchanged; a value close to zero strongly suppresses it.

The gate compares two clocks:

1. **Reorientation time** ($t_{reorient,i}$): how long the passer requires to turn sufficiently towards the target.
2. **Pressure time** ($t_{pressure}$): how long before the nearest defender reaches challenging distance.

If the pass can be played before pressure arrives, g_i remains close to one. If the required turn takes longer than the available pressure time, the option is suppressed.

The calculation proceeds in four stages: infer the passer's orientation, measure the turn required for each candidate, convert that turn into time, and compare it with the time available before pressure arrives. The following example applies each stage to two passing options available to Jude Bellingham.

A worked example

Two minutes into Real Madrid's match against Athletic Club, Bellingham receives the ball with his momentum carrying him back towards his own goal at approximately 4.0 m/s. His nearest defender is 4.4 m away and closing.

We compare two potential passes:

- **David Alaba**, the team-mate Bellingham ultimately selects, who lies close to his direction of travel.
- **Vinícius Júnior**, positioned further upfield and in almost the opposite direction.

Pitch control considers both passing lanes potentially available. The executability gate distinguishes whether Bellingham can realistically play them from his movement and orientation at that instant.

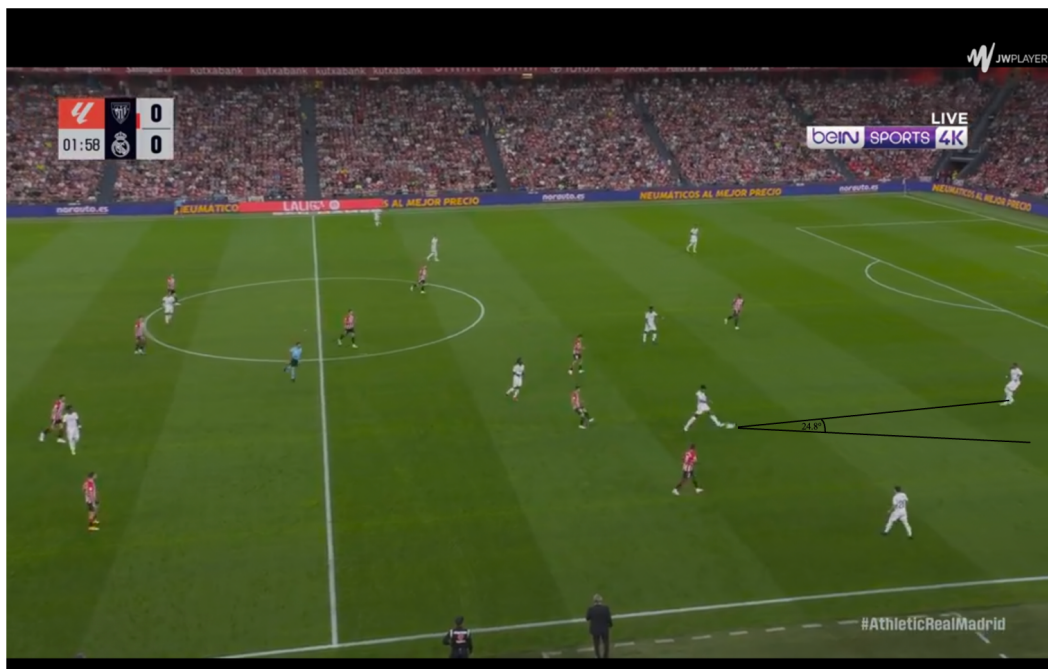


Figure 1. Bellingham is moving towards his own goal at 4.0 m/s. The lower ray shows his inferred facing direction from velocity, while the upper ray points towards Alaba. The angle between them is 24.8 degrees.

Inferring the passer's orientation

SkillCorner does not provide body-pose data, so the passer's facing direction must be inferred. The source and confidence of this estimate depend on the player's movement and recent reception history, as summarised below.

Passer at the pass frame	Facing taken from	Confidence
Moving, $v \geq 1.5$ m/s	Velocity direction	$\text{conf}(v) = \sigma(2.0(v - 2.5))$
Slow, with a matched incoming pass	Incoming pass direction, reversed 180°	0.70 if one-touch or holding the ball for $t \leq 0.7$ s; 0.35 if $0.7 < t \leq 1.5$ s; 0.10 beyond 1.5 s Where t is time on the ball before passing
Slow, with no matched incoming pass	Unobservable	0

For a player moving at $v \geq 1.5$ m/s, velocity direction is used as a proxy for facing. Confidence in that estimate increases with speed according to:

$$\text{conf}(v) = \sigma(2.0(v - 2.5)), \quad v \geq 1.5,$$

where σ denotes the sigmoid function and v is the passer's speed in metres per second.

For a slow-moving player with a matched incoming pass, orientation is inferred by reversing the direction from which the ball arrived. Confidence decreases as the time since reception increases because the player has more opportunity to turn after receiving. If neither movement nor a recent incoming pass provides a reliable estimate, orientation is treated as unknown and $\text{conf} = 0$. In that case, the gate does not suppress the passing option.

As Bellingham is moving at 4.0 m/s, his velocity vector supplies the facing estimate. Substituting his speed into the confidence equation gives:

$$\text{conf}(4.0) = \sigma(2.0(4.0 - 2.5)) = \sigma(3.0) = 0.95.$$

Bellingham's velocity therefore provides a high-confidence estimate of his orientation. With that orientation established, the next step is to measure how far he must turn towards each candidate.

Measuring the required turn

For each candidate i , the orientation gap is the angle between the passer's inferred facing direction and the direction from the passer to that candidate:

$$\text{gap}_i = \angle(\text{facing direction}, \mathbf{p}_i - \mathbf{p}_{\text{passer}}).$$

In this frame, the two orientation gaps are:

$$\text{gap}_{\text{Alaba}} = 24.8^\circ, \quad \text{gap}_{\text{Vini}} = 161.2^\circ.$$

The orientation gap is not yet the turn Bellingham must make. A player can pass across his body or somewhat behind him without first facing the target exactly, so the model allows a

111° turn-free range from the inferred facing direction. Only the angle beyond this threshold is counted:

$$\text{excess}_i = \max(0, \text{gap}_i - 111^\circ).$$

For Alaba, the 24.8° gap lies comfortably inside the turn-free range:

$$\text{excess}_{\text{Alaba}} = \max(0, 24.8^\circ - 111^\circ) = 0^\circ.$$

No reorientation is therefore required. Vinícius lies outside the turn-free range:

$$\text{excess}_{\text{Vini}} = \max(0, 161.2^\circ - 111^\circ) = 50.2^\circ = 0.877 \text{ radians}.$$

This excess angle, rather than the full 161.2° orientation gap, is used to calculate reorientation time.

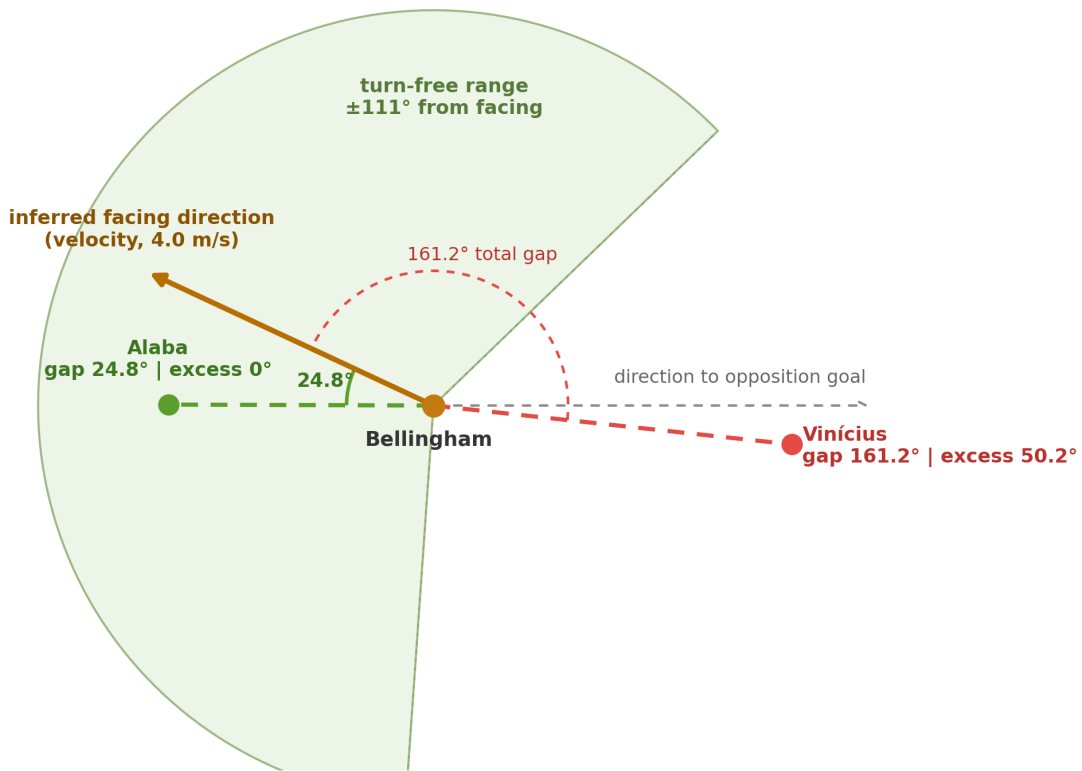


Figure 2. Alaba lies inside Bellingham's turn-free range and requires no reorientation. Vinícius lies 161.2 degrees from Bellingham's facing direction, leaving an excess turn of 50.2 degrees beyond the 111-degree threshold.

Converting the required turn into time

The excess angle tells us how far Bellingham must turn. To determine whether he can complete that turn before pressure arrives, it must now be converted into time. This depends on his estimated turning speed, which decreases as his running speed increases:

$$\omega(v) = \max\left(0.4, \frac{2.87}{1 + 1.14v}\right) \text{ rad/s}.$$

Substituting Bellingham's speed gives:

$$\omega(4.0) = \max\left(0.4, \frac{2.87}{1 + 1.14(4.0)}\right) = 0.52 \text{ rad/s} \approx 30^\circ/\text{s}.$$

Reorientation time combines the excess angle with this turn rate:

$$t_{\text{reorient},i} = \begin{cases} 0, & \text{excess}_i = 0, \\ 0.10 + \frac{\text{radians}(\text{excess}_i)}{\omega(v)}, & \text{excess}_i > 0. \end{cases}$$

The additional 0.10 seconds represents a fixed plant-and-strike cost once a turn is required. For Alaba:

$$t_{\text{reorient},\text{Alaba}} = 0 \text{ s}.$$

For Vinícius:

$$t_{\text{reorient},\text{Vini}} = 0.10 + \frac{\text{radians}(50.2^\circ)}{0.52} = 0.10 + \frac{0.877}{0.52} = 1.80 \text{ s}.$$

We now know the time required to produce each pass. The remaining question is how much time Bellingham has before the defender arrives.

The pressure clock

Pressure time estimates how long the nearest defender requires to reach a 1.5 m challenge radius around the passer. The movement model assumes acceleration of 7 m/s², up to a maximum speed of 5 m/s.

The nearest Athletic Club defender is 4.4 m from Bellingham and must cover approximately 2.9 m to reach the challenge radius. Under the movement model:

$$t_{\text{pressure}} = 0.93 \text{ s}.$$

If a defender is already inside the challenge radius, $t_{\text{pressure}} = 0$; times above 2.5 seconds are treated as comfortably open.

Combining the two clocks

Bellingham requires no reorientation to pass to Alaba, whereas the pass to Vinícius requires 1.80 seconds. Only 0.93 seconds are available before pressure arrives.

For an option requiring reorientation, the raw gate is:

$$g_{\text{raw},i} = \sigma\left(3.0(t_{\text{pressure}} - t_{\text{reorient},i})\right).$$

A positive timing margin produces a value near one; a negative margin produces a value near zero. For Vinícius:

$$g_{\text{raw},\text{Vini}} = \sigma(3.0(0.93 - 1.80)) = \sigma(-2.61) = 0.07.$$

Because Bellingham's orientation is inferred rather than directly observed, this raw result is moderated by the confidence weight. Options inside the turn-free range remain fully available:

$$g_i = \begin{cases} 1, & \text{excess}_i = 0, \\ 1 - \text{conf}(1 - g_{\text{raw},i}), & \text{excess}_i > 0. \end{cases}$$

For Vinícius:

$$g_{\text{Vini}} = 1 - 0.95(1 - 0.07) = 0.11.$$

For Alaba:

$$g_{\text{Alaba}} = 1, \quad \text{excess}_{\text{Alaba}} = 0^\circ.$$

The gate therefore leaves Alaba fully available but reduces the Vinícius option to 0.11, confirming that Bellingham is unlikely to reorient and strike that pass before pressure arrives.

Parameter summary

The parameter values used in the calculation are summarised below.

Component	Value	Basis
Base turn rate	2.87 rad/s	Fitted from tracking
Momentum penalty	1.14	Fitted from tracking
Turn-free half-angle	111°	Data-anchored
Half-confidence speed	2.5 m/s	Data-anchored
Moving-orientation threshold	1.5 m/s	Fixed
Confidence steepness	2.0	Fixed
Gate steepness	3.0	Fixed
Challenge radius	1.5 m	Fixed
Open-time cap	2.5 s	Fixed
Plant-and-strike cost	0.10 s	Fixed
Minimum turn rate	0.4 rad/s	Fixed

The turn-rate curve is fitted from observed reorientation rates in the tracking data. The turn-free angle and half-confidence speed are anchored in the data, while the remaining constants were sensitivity-tested rather than estimated directly.

3.4: V_{cmp} , V_{turnover} : Value of the arrival location

Each passing option is evaluated conditionally on its outcome: what the position is worth if the pass arrives, and what it costs if it does not.

V_{cmp} : value if the pass is completed

If a pass reaches receiver i , we estimate the attacking value of the resulting state from the open-play xG subsequently generated during that possession. Shots closer in time to the pass receive more credit through an exponential discount:

$$V_{cmp} = \sum_{\substack{s \in \text{shots}(P) \\ f_s \geq f}} xG(s) e^{-\Delta t_s / \tau = 30}$$

Where Δt_s is the time between the pass and shot s

The discount acts as a simple credit-assignment mechanism: a shot immediately after the pass retains almost all of its xG, while a shot 30 seconds later receives approximately $e^{-1} \approx 0.37$ of its original value.

A gradient-boosted model predicts this target using features describing the reception state: the receiver's location and geometry relative to goal, nearby defensive pressure, the number of goal-side defenders and the attacking support ahead of the ball.

 $V_{turnover}$: cost if the pass fails

The turnover head is constructed symmetrically. If the pass is lost and the opposition gains possession, we measure the discounted open-play xG generated in the immediately following opposition possession and assign it a negative value:

$$V_{turnover} = - \sum_{s \in \text{shots}(Q)} xG(s) e^{-\Delta t_s / \tau}$$

This allows an option to reflect both its potential attacking reward and the risk associated with losing the ball.

Combining the completion probability with the two value states gives the expected value of passing to candidate i : $\underbrace{P_{cmp,i} \cdot V_{cmp,i}}_{\text{upside reward}} + \underbrace{(1 - P_{cmp,i}) \cdot V_{turnover,i}}_{\text{downside risk}}$

Evaluation

Both value heads are trained using gradient-boosted regression and evaluated using by-game cross-validation. Because the discounted xG target is zero for most passes, we report the ROC-AUC for whether the relevant possession eventually produces a shot, yielding ROC-AUCs of 0.61 for V_{cmp} and 0.55 for $V_{turnover}$ respectively. These results are modest, reflecting the sparsity of the target: only around 10% of open-play possessions end in a shot. This limitation is discussed further in Section 7.

3.4.1: Expected Goals model

SkillCorner does not provide xG values, so we train a separate expected-goals model using StatsBomb open data on 36,508 open-play shots from the Big Five European leagues in 2015/16 (Hudl StatsBomb).

Because this model must later be applied to SkillCorner tracking data, only features that can be reproduced consistently in both datasets are used. These comprise distance and angle to goal, the shot's location, a header flag, distance to the goalkeeper, the goalkeeper's distance to goal, local defensive pressure, defenders occupying the direct shooting lane, and whether the goal is effectively open.

An XGBoost model achieves an out-of-sample ROC-AUC of **0.818**, close to the **0.826** achieved by StatsBomb's own xG values on the same shots. StatsBomb and SkillCorner coordinates are first rescaled to a common 105×68 m pitch before the same features are calculated.

3.5 Pass-into-space

So far, every passing option has been evaluated at the receiver's location when the pass is made. This works for passes to feet, but can misprice passes played ahead of a runner.

Consider a striker running beyond the defensive line. The lane to his current position may be blocked and that position may still be relatively low value, even though a pass into the space ahead of him is both easier to complete and more threatening. A feet-only model can therefore systematically undervalue through-balls.

Adding an in-space target

For a candidate i making a tagged off-ball run, we evaluate a second target alongside his current position: a projected **rendezvous point** Z_i where the runner and ball could meet.

The candidate's passing value is then the better of the two available targets:

$$\max\{xPV_i(feet), xPV_i(Z_i)\}$$

This treats Z_i as an additional way of reaching the same receiver rather than as a separate candidate.

Locating Z_i

Let p_i and v_i denote the receiver's position and velocity at the pass frame, and p_p the passer's position. Assuming the receiver maintains his current direction and speed over the short flight of the ball, his projected position after t seconds is $p_i + v_i t$.

The distance the ball must travel to meet him at that time is therefore

$$D(t) = \|(p_i + v_i t) - p_p\|$$

Ball speed is estimated using the same distance-dependent relationship as the pitch-control model: $s(L) = 5.0 + 0.62 \cdot L$, capped at 28 m/s, where L is pass length.

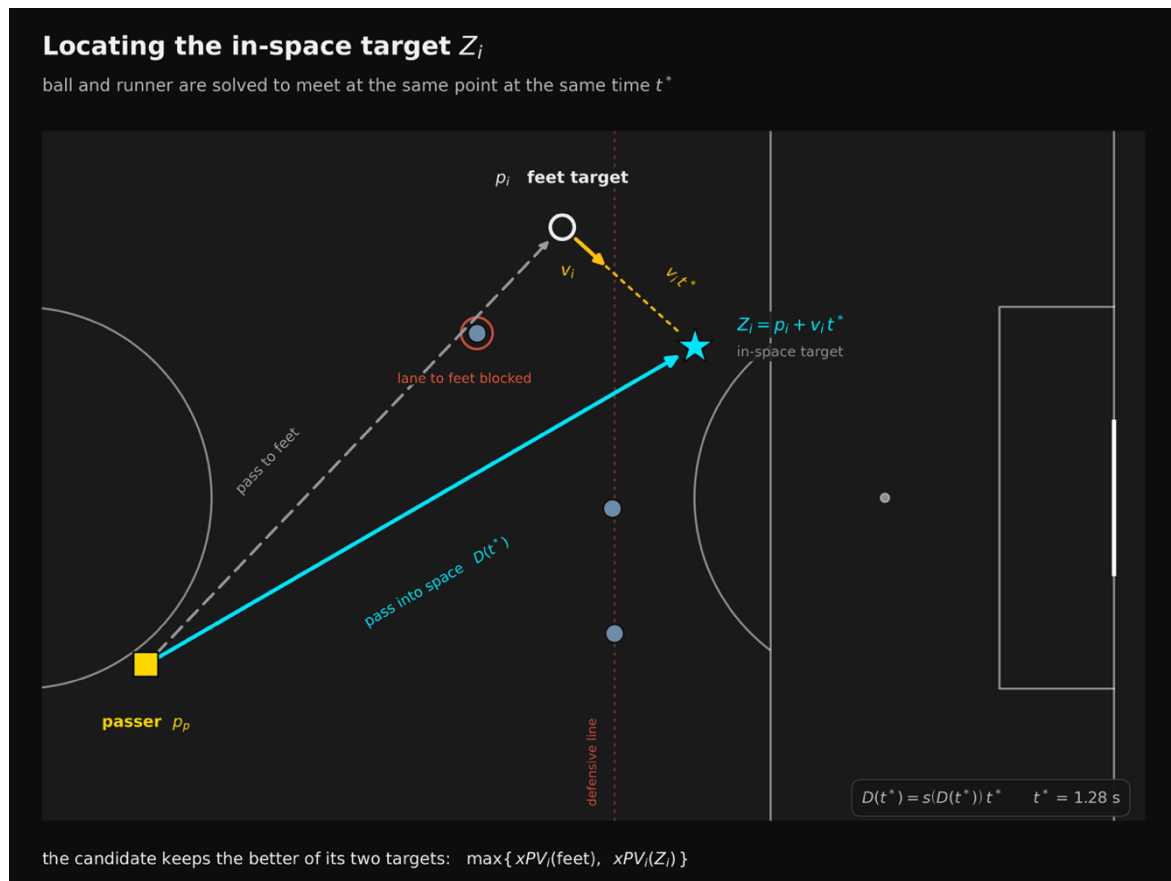
We solve for the earliest positive time t^* at which the projected runner and ball can arrive at the same location:

$$D(t^*) = s(D(t^*)) t^*$$

The resulting in-space target is:

$$Z_i = p_i + v_i t^*$$

The distance ahead therefore adapts naturally to the situation rather than being fixed by hand. A fast runner and a longer pass produce a deeper rendezvous point, while a slowly moving receiver is met closer to his current position.



Re-valuing the option at Z_i

The feet-based value is not simply transferred to the projected destination. Instead, the full passing option is recomputed at Z_i . The completion model evaluates the passing lane and defensive pressure toward the new target, the executability gate evaluates whether the passer can strike the ball in that direction, and both value heads evaluate the state created at the projected arrival location:

$$P_{cmp}(Z_i) = g(Z_i) P_{model}(Z_i)$$

$$xPV_i(Z_i) = P_{cmp}(Z_i)V_{cmp}(Z_i) + (1 - P_{cmp}(Z_i))V_{turnover}(Z_i)$$

This is important: a pass into space may differ from a pass to feet not only because it arrives further upfield, but because the passing lane, interception risk and required body orientation are also different.

Example: Kroos pass into space for Brahim Díaz



T. Kroos → Brahim Díaz · Real Madrid CF v Granada CF, 25:33 · successful

the same option, priced at the receiver's feet and at the projected in-space target Z_i

background = pitch control · gold ■ passer · amber ○ runner + v_i · cyan ★ Z_i · white ○ actual reception · red | offside line



	P_model	gate g	P_cmp	V ⁺	V ⁻	xPV
feet	0.59	1.00	0.59	0.035	-0.003	+0.0194
in-space	0.83	1.00	0.83	0.112	-0.002	+0.0925

pricing the same pass at Z_i instead of at his feet moves it up 0.0731 xPV · Z_i lands 0.8 m from where the ball arrived (his feet were 6.9 m away)

The model recognises a pass to Brahim Díaz’s feet is far more challenging and less effective than the option into space.

Validation

Across 13,507 passes classified as into-space options, the model predicted a mean completion probability of 79.8%, compared with an observed completion rate of 77.8%, with a Brier score of 0.1339. This provides an aggregate check that evaluating completion at the projected rendezvous point produces plausible probability estimates.

We also sanity-checked the construction on hand-reviewed successful through-balls and overlaps. In these examples, evaluating the pass at Z_i generally produced more plausible completion and value estimates than feet-only pricing.

The rendezvous construction remains an approximation - particularly because it assumes constant receiver velocity - and is discussed further in the limitations section.

4. Metrics

Because xPV assigns a value to every available passing option, it can evaluate not only the pass that was played but also the quality of the choice itself. Let xPV_c denote the value of the chosen option and xPV the π -weighted value of the decision state.

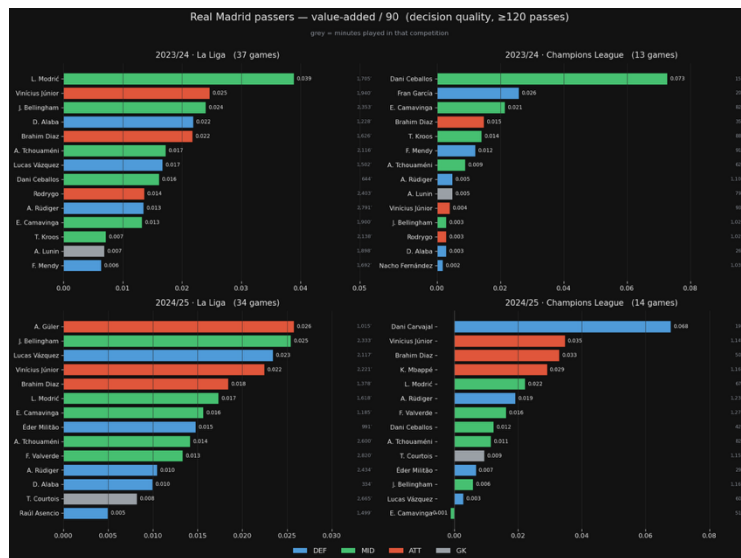
Metric	Definition	Interpretation
xPV added	$xPV_c - xPV$	Value gained relative to the options normally chosen from that state
Chose-best %	$\mathbb{1} \left[xPV_c = \max_i xPV_i \right]$	How often the highest-valued pass was selected
Decision percentile	rank of xPV_c	Where the chosen pass ranked among available options
Regret	$\max_i xPV_i - xPV_c$	Value left on the table
Uniqueness	$\mathbb{1} \left[c \neq \arg \max_i \pi_i \right]$	How often the choice differed from the model's expected receiver

For comparison with existing pass-value metrics, we also report **chosen-xPV**, the expected value of the pass actually played.

5. Player Results

We apply the decision metrics across the 98 Real Madrid matches, separating the 2023/24 and 2024/25 La Liga and Champions League samples. To avoid rankings being driven by very small samples, only players with at least 120 passes in a competition-season are shown.

5.1 Value added and decision quality

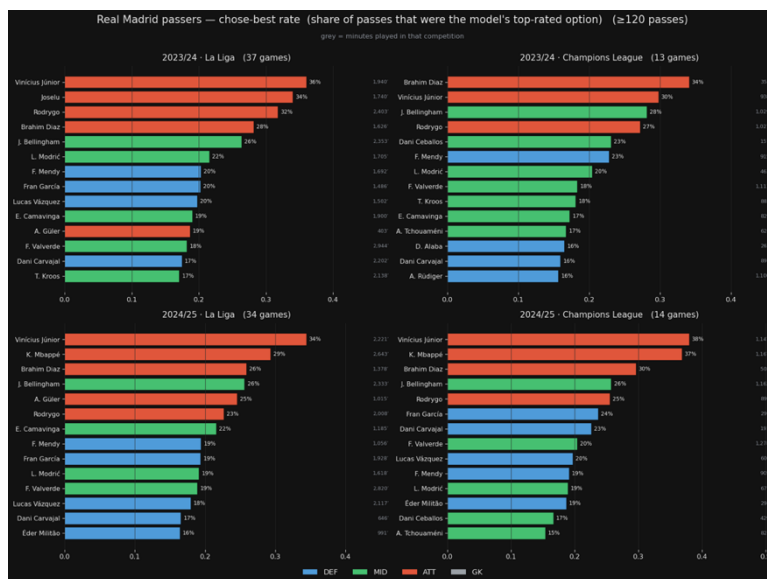


The results highlight players who consistently chose valuable passing options, rather than simply those who generated the most threat.

Modrić stands out particularly strongly on xPV added, while Vinicius Jr, Bellingham and Brahim Díaz also rank highly across multiple samples. These rankings capture performance relative to the options available at each decision: a player is rewarded for selecting a strong option from a difficult situation, rather than simply for occupying positions from which high-value passes are easier to make.

Decision percentile provides a complementary view. Rather than measuring the absolute value added by a choice, it asks where the chosen pass ranked among the alternatives available at that moment. This makes it less sensitive to the overall value of the situation and provides a more direct measure of option selection.

5.2 Chose-best and Uniqueness



Chose-best measures the strictest version of decision quality: how often the player selected the model's highest-valued passing option. Decision percentile is therefore generally the more forgiving measure, since choosing the second-best of many similar options can still represent a strong decision.

Uniqueness captures something different. It measures how often a player chooses a receiver other than the one the selection model considered most likely. Modrić, Kroos and, interestingly, Courtois appear among the more unpredictable passers in the sample. This should not be interpreted as a measure of quality: an unexpected pass may be either excellent or poor. Instead, uniqueness describes **how predictable a player's passing choices are**, providing a separate behavioural dimension alongside the value-based decision metrics.

6. Benchmarking against xT

Expected Threat is the natural public benchmark for xPV. It assigns every pitch zone a value - the probability that possessing the ball there leads to a goal - and scores a pass by the threat between its start and end locations (Singh, 2018):

$$xT\text{-added} = xT(s_{\text{end}}) - xT(s_{\text{origin}})$$

Beyond ignoring the alternatives a passer had, another limitation of the standard xT is that it contains no explicit downside for a failed pass. A completed forward pass earns the threat gained, while an unsuccessful pass typically receives zero regardless of the value of possession lost.

We therefore construct two adjusted versions. **Context-aware xT** applies a negative xT penalty when a failed pass results in genuine opposition control (a clean interception, sustained opponent possession or an opponent possession ending in a shot -rather than

penalising balls that are immediately recovered by the passing team). **xT ex-cross** additionally removes crosses, which can receive a large xT increase simply from entering a high-value penalty-box zone and are more naturally captured by expected-assist measures.

As a completion-weighted benchmark, we also use Shaw's EPV formulation (Shaw, 2020), which weights the threat at each end of the pass by pitch control:

$$PC(R_f) \cdot xT(R_f) - PC(R_i) \cdot xT(R_i)$$

6.1 What the metrics reveal

Across the 25 Real Madrid players with at least 500 minutes and 300 open-play passes, we compare the player-level decision metrics with these existing pass-value measures.

Decision-metric agreement across pass-value models

Only xPV/90 correlates with whether the passer picked the best available option; the xT family tracks value added but not selection.

DECISION METRIC	xPV/90	xT	ctx-xT	xT ex-cross	Laurie Shaw EPV
value-added	+0.76 ***	+0.75 ***	+0.76 ***	+0.61 **	+0.45 *
chose-best	+0.41 *	+0.09	+0.13	-0.21	-0.01
percentile	+0.45 *	+0.12	+0.17	-0.19	-0.01
uniqueness	-0.10	+0.05	+0.10	+0.17	+0.28

 correlation
 n = 25 (Laurie Shaw EPV n = 21) · * p<.05, ** p<.01, *** p<.001

xPV value-added shows clear positive association with the threat-based benchmarks, with correlations ranging from approximately $r = 0.45$ to 0.76 . All three xT-based correlations are statistically significant at ($p < .01$). This provides encouraging evidence that players who add value relative to the passing options available to them also tend to generate more threat by conventional pass-value measures.

The pattern is very different for **chose-best** and **decision percentile**. Both correlate moderately with chosen-xPV per 90 ($r \approx 0.4$), but show little association with xT, context-aware xT, xT ex-cross or Shaw's EPV.

This suggests that the decision metrics may be capturing a different aspect of passing performance. Traditional threat metrics are strongly influenced by where a player receives and moves the ball; chose-best and percentile instead ask how highly the selected pass ranked among the alternatives available in that particular situation.

Whether this represents a stable player skill or simply noise in the current sample remains an open question. A key next step is therefore to test whether players who consistently make higher-ranked decisions also produce better realised outcomes over larger samples.

7. Limitations and Next Steps

The framework evaluates passing decisions against the alternatives available at the same moment, but several parts of the current implementation remain approximate.

7.1 The value target is the main bottleneck

The clearest limitation is the target used by V_{cmp} and V_{turnover} . Both are learned from discounted xG generated later in the possession, but only around 10% of open-play possessions produce a shot. Most labels are therefore zero, making it difficult to distinguish genuinely low-value states from useful circulation or retention that does not lead directly to a shot.

This is reflected in the modest shot-occurrence AUCs of 0.61 and 0.55. More importantly, a 24-configuration hyperparameter sweep changed V_{cmp} AUC by no more than 0.004. The limited improvement from tuning suggests that the value target, rather than model capacity, is the main bottleneck. Kroos provides an intuitive example of the problem: passes that retain possession, reset an attack or draw pressure can be valuable without sitting directly upstream of a shot, yet the current target gives them little credit reflected in his lower value added metrics.

7.2 Passes into space remain approximate

The rendezvous extension improves on valuing every option at the receiver's feet, but Z_i is still a simplified representation of a continuous passing opportunity. It assumes that the receiver maintains his current speed and direction during the ball's flight.

This works well for many established runs but can misprice cases where a slowly moving player subsequently accelerates sharply into space, or where the runner changes direction after the pass frame. The estimated rendezvous point may then be too shallow, too deep or on the wrong line.

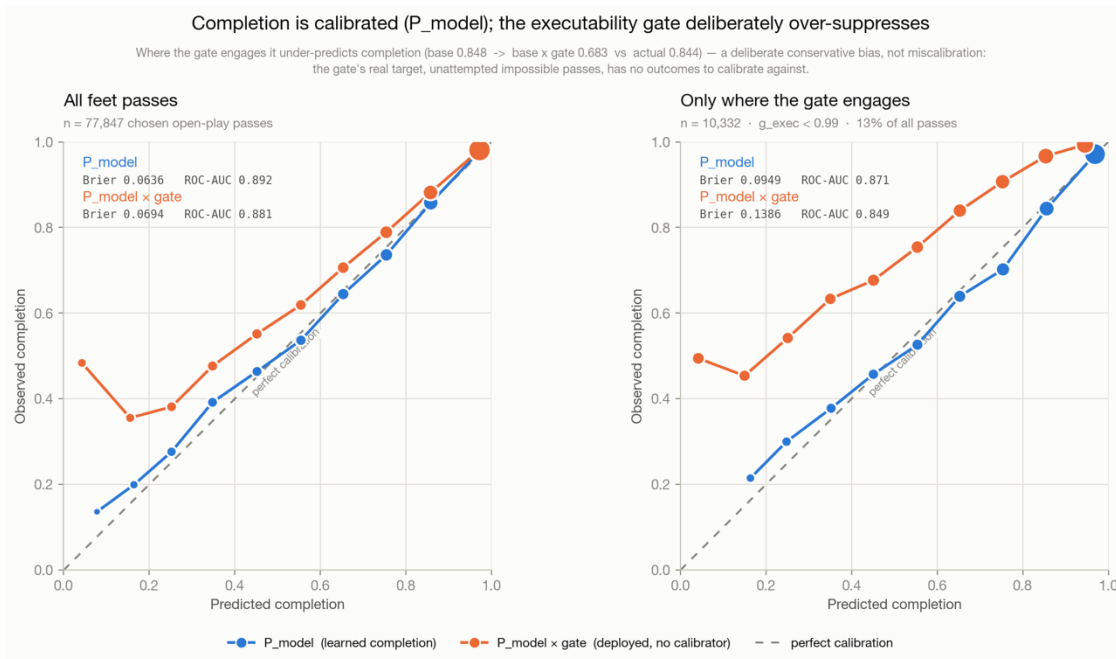
A richer version could evaluate a small surface of plausible destination points - potentially incorporating pitch control - rather than committing to a single Z_i . The current implementation also depends on SkillCorner's off-ball-run tags, while the learned completion model is trained at observed player locations but applied at the projected destination.

7.3 Executability depends on inferred orientation

SkillCorner does not provide body pose, so the executability gate infers orientation from player movement and, for slow-moving recent receivers, the incoming pass. Confidence weighting makes the gate conservative when this orientation is uncertain, but the underlying facing direction remains an approximation.

There is also a calibration trade-off. P_{model} is trained on passes that were actually attempted and is reasonably calibrated on those observations. Multiplying by the gate reduces predicted completion, particularly in situations where g engages, and therefore worsens calibration against observed attempted passes.

This does not necessarily mean that the gate is serving the wrong purpose: it is intended to suppress counterfactual options that may have been physically difficult or impossible to attempt, and such options have no observed completion outcomes against which to calibrate. Nevertheless, separating calibrated delivery probability and counterfactual feasibility more cleanly is an important area for improvement.



7.4 Evaluating decision making *at the moment of the pass*

The current framework evaluates which team-mate the player selected, conditional on a pass being made at that moment. It does not assess whether passing was preferable to shooting, carrying the ball or retaining possession, nor whether the player should have passed earlier or later. Consequently, it may overlook better passing opportunities that existed before the eventual pass frame.

A full-scale EPV framework could address this by valuing multiple actions continuously across successive game states. This would allow the model to compare passing with shooting or carrying, while also evaluating whether the player acted at the most valuable moment.

7.5 Priorities for future work

The highest-priority extension is a richer state-value target. A bootstrapped formulation:

$$V(s) = r + \gamma V(s'),$$

could propagate value through successive possession states rather than relying entirely on whether the possession eventually produces a shot. This should provide more signal for circulation, retention and progression that the current target often misses.

The next priority is improving the modelling of counterfactual passing options, particularly body orientation, feasibility and the destination of passes into space. Longer term, xPV could be embedded within a full EPV framework that evaluates actions continuously rather than only at pass moments.

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